8.4 Answers: word problems

← Unit overview · Practice

For the parent: in this unit, the steps matter more than the number. If he used the five steps and slipped on one calculation, that is a good result. Say so.

Set A: Just plan it

1. **(b) 3 × 7,50.** Three times the same price. (= €22,50)
2. **(c) 96 : 32.** How many 32s fit in 96. (= 3 buses)
3. **(a) 15 + 8.** She gets more. (= €23)
4. **(b) 12 − 2,5.** Cutting off makes it shorter. (= 9,5 m)
5. **(c) 24 : 6.** Fair sharing is dividing. (= 4 each)
6. **(a) and (b) are both right.** Half a kilo is half the price: 12 : 2 and 0,5 × 12 both give €6. That's the surprise.
7. **(b) count on.** 10:40 → 11:40 is 1 hour, → 12:05 is 25 more minutes. (= 1 hour 25 minutes)
8. **(b) 4 × 7.** Rows of equal size. (= 28 chairs)

Set B: One step

1. **€5.** 12 − 7.
2. **9 stroopwafels.** 36 : 4.
3. **€24.** 3 × €8.
4. **22 seats.** 50 − 28.
5. **16:15.** 14:30 + 1 h = 15:30; + 45 min = 16:15.
6. **5 packs.** 30 : 6.
7. **8 km.** 4 there + 4 back.
8. **€6.** 3 × €2.

Set C: Two steps

1. **€2,50.** 3 × €2,50 = €7,50; €10 − €7,50.
2. **7 groups.** 27 : 4 = 6 groups with 3 children left over; they need a group too (so 6 groups of 4 and one of 3).
3. **4 slices each, 4 left over.** 3 × 8 = 24 slices; 24 : 5 = 4 rest 4.
4. **€170.** 2 × €45 = €90; 2 × €40 = €80; €90 + €80.
5. **8 weeks.** €3,50 × 2 = €7 per 2 weeks; €28 is 4 × €7, so 8 weeks.
6. **€7,80.** 4 × €1,25 = €5,00; + €2,80.
7. **6 glasses.** 1,5 l = 1 500 ml; 1 500 : 250 = 6.
8. **12:20.** 9:45 + 2 h = 11:45; + 35 min = 12:20.
9. **12 mm.** 12 + 8 + 16 = 36; 36 : 3.
10. **7 cookies.** 10 − 3.

Set D: Several steps

1. **150 children.** 8 × 25 = 200; ¾ of 200 = 150.
2. **3 buses.** 150 : 50 = 3 exactly.
3. **1,5 litres; buy 2 tins.** Wall = 3 × 4 = 12 m²; 12 : 8 = 1,5 l; one tin isn't enough, so 2.
4. **€40 per month.** 240 : 6.
5. **12 minutes.** Cross out "12 years old". 15 km in 60 min → 3 km in 12 min (divide both by 5).
6. **€95.** 4 × €22,50 = €90; + €5.
7. **€7,50 profit.** Buys for 50 × €0,20 = €10; sells for 50 × €0,35 = €17,50; €17,50 − €10.
8. **270 children.** 10% = 60, 40% = 240, 5% = 30; 240 + 30.
9. **9 eggs.** 12 people is 3 × 4 people; 3 × 3 eggs.

10. **60 pencils.** 5 × 12.

Set E: Tables and timetables

1. **Every 30 minutes.** 8:05, 8:35, 9:05.
2. **36 minutes.** 8:05 → 8:41.
3. **The 8:42 bus.** It reaches the Ziekenhuis at 8:50. The 9:12 arrives at 9:20, too late.
4. **29 minutes.** 8:12 → 8:41.
5. **Wednesday** (210 visitors).
6. **650 visitors.** 120 + 95 + 210 + 80 + 145.
7. **130 visitors per day.** 650 : 5.
8. **130 more.** 210 − 80.
9. **120 visitors.** Read it from the table.

Puzzle corner

1. **Day 8.** After 7 days and nights it is at 7 m. On day 8 it climbs 3 m to 10 m and is out before night comes. (Most people say day 10 at first.)
2. **6 chickens and 4 cows.** If all 10 were chickens there'd be 20 legs. Each cow adds 2 extra legs; 28 − 20 = 8 extra, so 4 cows. Check: 6 × 2 + 4 × 4 = 12 + 16 = 28.
3. **Any problem that works.** Check it together: does the story really give €4,50?